

$$13) E = 110 \text{ GPa} \quad \sigma_y = 500 \text{ MPa}$$

$$\frac{F}{A} = E \epsilon$$

$$\frac{F}{\frac{\pi d^2}{4}} = E \frac{\delta}{L_0}$$

$$\frac{F L_0}{E \delta} = \frac{\pi d^2}{4}$$

$$d^2 = \frac{4 F L_0}{\pi E \delta}$$

$$d^2 = \frac{4 (6660) (380 \times 10^{-3})}{\pi (110 \times 10^9) (0.5 \times 10^{-3})}$$

$$d^2 = 5.858753891 \times 10^{-5}$$

$$d = 7.554249728 \times 10^{-3} \text{ m}$$

$$\approx 7.65 \text{ mm}$$

$$14a) \sigma_y = E \epsilon$$

$$\frac{F}{A} = \frac{E \delta}{L_0}$$

$$\delta = \frac{F L_0}{A E}$$

$$\delta = \frac{14 \times 10^3 \times 20 \times 10^{-2}}{(10 \times 10^{-3})^2 (70 \times 10^9)}$$

$$= 4 \times 10^{-4} \text{ m}$$

$$= 0.4 \text{ mm}$$

$$b) \sigma_y = \frac{14 \times 10^3}{(10 \times 10^{-3})^2}$$

$$= 140 \text{ MPa}$$

At $\sigma = 200 \text{ MPa}$ and $\sigma = 220 \text{ MPa}$, $\sigma > \sigma_y$,

$$\sigma_T = K \epsilon_T^n$$

$$\ln \sigma_T = \ln K + n \ln \epsilon_T$$

$$n \ln \epsilon_T = \ln \sigma_T - \ln K$$

$$n + \frac{\ln K}{\ln \epsilon_T} = \frac{\ln \sigma_T}{\ln \epsilon_T}$$

14b) When $\sigma = 200 \text{ MPa}$, $\epsilon = 0.05$

$$\begin{aligned}\sigma_T &= 200(1+0.05) & \epsilon_T &= \ln(1+0.05) \\ &= 210 \text{ MPa} & &= 0.04879016417\end{aligned}$$

When $\sigma = 220 \text{ MPa}$, $\epsilon = 0.10$,

$$\begin{aligned}\sigma_T &= 220(1+0.1) & \epsilon_T &= \ln(1+0.1) \\ &= 242 \text{ MPa} & &= 0.0953101798\end{aligned}$$

Sub $\sigma_T = 200$, $\epsilon_T = 0.05$,

$$n + \frac{\ln K}{\ln 0.049} = \frac{\ln 210}{\ln 0.049} - (1)$$

Sub $\sigma_T = 220$, $\epsilon_T = 0.10$,

$$n + \frac{\ln K}{\ln 0.095} = \frac{\ln 242}{\ln 0.095} - (2)$$

Solving,

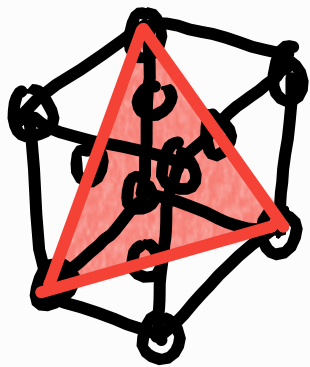
$$n = 0.2118108204$$

$$\ln K = 5.986824192$$

$$\therefore n \approx 0.212$$

15) Close packed directions:

For FCC, the closed packed directions have the highest linear density.



Slip direction: $\langle 110 \rangle$

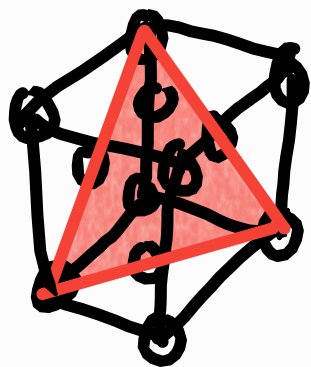
Linear density = 1

The slip planes are $\{111\}$

$\therefore$ The slip system for FCC is

$\langle 110 \rangle \{111\}$

16)



In the FCC cell,
there are 3 directions
in $\langle 110 \rangle$ for the plane
(111):

$$[01\bar{1}]$$

$$[\bar{1}01]$$

$$[1\bar{1}0]$$

There are 4 different $\{111\}$ planes,
with each containing 3 different
directions:

$$(111)[\bar{1}01], (111)[\bar{1}01], (111)[\bar{1}01]$$

$$(\bar{1}11)[101], (\bar{1}11)[110], (\bar{1}11)[01\bar{1}]$$

$$(1\bar{1}1)[011], (1\bar{1}1)[110], (1\bar{1}1)[\bar{1}01]$$

$$(11\bar{1})[011], (11\bar{1})[101], (11\bar{1})[\bar{1}10]$$